Problem Set 3

Problem 1. Prove that if n, m are positive integers, then $\gcd(2^{2n+1}-1, 2^m+1) = 1$.

Problem 2. Let a, b, c be the lengths of the sides of a triangle ABC with inradius r and area A. Compute

$$\frac{\cos A - \cos B}{A - rc} + \frac{\cos B - \cos C}{A - ra} + \frac{\cos C - \cos A}{A - rb}.$$

Problem 3. How many ways are there to weigh 31 grams with a balance if we have 7 weights of one gram, 5 of two grams, and 6 of five grams, respectively?

Problem 4. Let $a > b > c > d > 0$ be positive real numbers such that $ad = bc$. Show that

$$\frac{\sqrt{a} - \sqrt{b}}{\sqrt{c} - \sqrt{d}} = \frac{\sqrt{a-b}}{\sqrt{c-d}}.$$

Problem 5. Find all triples (x, y, z) of real numbers such that

$$\sqrt[6x-y^2-7]{\sqrt[6y-z^2-7]{\sqrt[6z-x^2-7]{728x+729y+730z}}}$$

is a positive integer, and determine its value.

Problem 6. Let n be a positive integer. Compute

$$\frac{1}{2}\binom{n+1}{n} + \frac{1}{4}\binom{n+2}{n} + \cdots + \frac{1}{2^{n-1}}\binom{2n-1}{n} + \frac{1}{2^n}\binom{2n}{n}.$$

Conjunto de Problemas 3

Problema 1. Demuestre que si n, m son enteros positivos, entonces $\gcd(2^{2n+1}-1, 2^m+1) = 1$.

Problema 2. Sean a, b, c las longitudes de los lados de un triángulo ABC con inradio r y área A. Calcule

$$\frac{\cos A - \cos B}{A - rc} + \frac{\cos B - \cos C}{A - ra} + \frac{\cos C - \cos A}{A - rb}.$$

Problema 3. ¿De cuántas formas se pueden pesar 31 gramos con una balanza si tenemos 7 pesas de un gramo, 5 de dos gramos y 6 de cinco gramos, respectivamente?

Problema 4. Sean $a > b > c > d > 0$ números reales positivos tales que $ad = bc$. Demuestre que

$$\frac{\sqrt{a} - \sqrt{b}}{\sqrt{c} - \sqrt{d}} = \frac{\sqrt{a-b}}{\sqrt{c-d}}.$$

Problema 5. Encuentre todas las ternas (x, y, z) de números reales tales que

$$\sqrt[6x-y^2-7]{\sqrt[6y-z^2-7]{\sqrt[6z-x^2-7]{728x+729y+730z}}}$$

sea un entero positivo, y determine su valor.

Problema 6. Sea n un entero positivo. Calcule

$$\frac{1}{2}\binom{n+1}{n} + \frac{1}{4}\binom{n+2}{n} + \cdots + \frac{1}{2^{n-1}}\binom{2n-1}{n} + \frac{1}{2^n}\binom{2n}{n}.$$